

Foundations of Biomedical Informatics
Quiz - 2

Date - 27 January 2025
Max. Marks - 100

Time - 90 minutes
Name:

SECTION - A

Multiple Choice Questions (MCQs)

15*4=60

1. Which of the following best represents a Learning Health System?
A. A hospital that stores patient records digitally
B. A system that improves continuously using routine care data
C. A system that follows fixed rules
D. A system that avoids uncertainty
2. Why is standardization important in healthcare data?
A. To reduce hospital staff
B. To ensure shared understanding across systems
C. To increase computation speed
D. To store more data
3. Which transformation best captures the goal of biomedical informatics?
A. Data → storage → retrieval
B. Data → machine → output
C. Data → information → knowledge → decision
D. Measurement → automation → prediction
4. A clinical alert system generates many warnings. Doctors start ignoring most of them, including important ones. Which informatics issue is being illustrated?
A. Poor data collection
B. Human interoperability failure
C. Lack of standardization
D. Insufficient data volume
5. An antibiotic guideline reduces infections initially, but resistance increases after repeated use. Which principle best explains this outcome?
A. Data scarcity
B. Closed-world assumption
C. Learning health system failure
D. Technical interoperability
6. Why does collecting more data not always improve healthcare decisions?
A. Data collection is slow
B. Bias and misinterpretation can still exist
C. Storage is expensive
D. Clinicians resist technology
7. Why is standardization essential in healthcare data exchange?
A. To reduce data volume
B. To speed up networks
C. To ensure consistent interpretation
D. To automate decisions
8. A patient's temperature is recorded as 38°C. Which additional context is MOST critical?
A. Hospital name

B. Measurement method

- C. Storage format
- D. Network latency

9. **Assertion (A):** Healthcare data cannot be interpreted correctly without context.
Reason (R): The meaning of biomedical data depends on who, when, and how it was collected.
A. Both Assertion and Reason are true, and Reason explains Assertion
B. Both are true, but Reason does NOT explain Assertion
C. Assertion is true, Reason is false
D. Assertion is false, Reason is true
10. **Assertion (A):** Large volumes of data guarantee better healthcare decisions.
Reason (R): Bias and noise can persist even in large datasets.
A. Both Assertion and Reason are true, and Reason explains Assertion
B. Both are true, but Reason does NOT explain Assertion
C. Assertion is true, Reason is false
D. Assertion is false, Reason is true
11. Assertion (A): De-identified data can still pose privacy risks.
Reason (R): Linking multiple datasets can re-identify individuals.
A. Both Assertion and Reason are true, and Reason explains Assertion
B. Both are true, but Reason does NOT explain Assertion
C. Assertion is true, Reason is false
D. Assertion is false, Reason is true
12. Assertion (A): HIPAA allows data use without consent in some cases.
Reason (R): Healthcare operations require continuous data analysis.
A. Both Assertion and Reason are true, and Reason explains Assertion
B. Both are true, but Reason does NOT explain Assertion
C. Assertion is true, Reason is false
D. Assertion is false, Reason is true
- Write True or False:**
13. A Learning Health System improves only when new technology is added. **F**
14. The same clinical measurement can have different interpretations for different patients. **T**
15. Health insurance providers influence healthcare outcomes only through financial reimbursement and have no impact on clinical decision-making. **F**

SECTION-B

Long Answer Questions

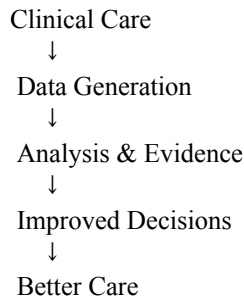
4*10 = 40

1. Describe the Learning Health System (LHS) framework.(3) What are feedback loops?(2) Explain how continuous feedback loops, and real-world data integration distinguish LHS from traditional healthcare systems (5).

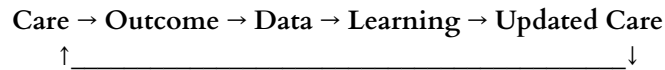
Ans.

LEARNING HEALTH SYSTEM (3)

A **Learning Health System** is a framework in which routine clinical care continuously generates data, this data is analyzed to create evidence, and the evidence is fed back into practice to improve patient care over time. The goal of a Learning Health System is to provide the best possible care at lower cost, while enabling continuous improvement over time through learning from real-world clinical practice.

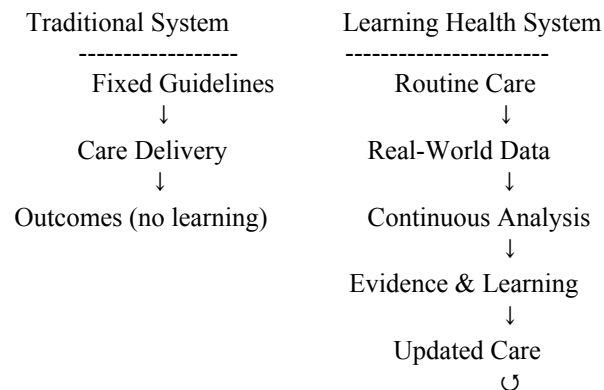


FEEDBACK LOOPS (2)



- Feedback loops are cyclical processes where outcomes of healthcare delivery produce data (1)
- This data is analyzed and used to modify future clinical decisions, guidelines, or policies (1)

Explain how continuous feedback loops, and real-world data integration distinguish LHS from traditional healthcare systems (5).



- Traditional healthcare systems rely on static models and fixed guidelines with infrequent updates (1)
- Learning Health Systems expect adaptation and continuous feedback loops (1)
- LHS integrates real-world data from routine care, not only controlled experiments (1)

- Models are updated continuously using real-world data, enabling ongoing learning (1)
- Policies are designed assuming unknown unknowns, focusing on resilience rather than certainty (1)

2. **Describe Eroom's Law and explain its significance for biomedical informatics research.(4) How does it influence the way we design data-driven healthcare systems and evaluate technological progress? (6)**

Ans.. Describe Eroom's Law and explain its significance for biomedical informatics research.

- Eroom's Law states that the cost and time required to develop new drugs increase over time, despite advances in science and technology (1)
- It represents the inverse of Moore's Law, where productivity decreases rather than increases (1)
- This trend highlights that biomedical innovation does not automatically scale with technological progress (1)
- It is significant for biomedical informatics because it exposes system-level inefficiencies in how biomedical data, models, and knowledge are used (1)

How does it influence the way we design data-driven healthcare systems and evaluate technological progress? (6)

- Encourages moving away from one-time, static models toward learning systems that continuously adapt (1)
- Motivates the use of real-world clinical data instead of relying only on controlled trials (1)
- Supports continuous feedback loops, where care outcomes inform future decisions (1)
- Shifts the evaluation of technology from raw computational power to measurable improvements in care quality and efficiency (1)
- Emphasizes systems thinking, recognizing healthcare as a complex, adaptive system rather than a closed world (1)
- Reinforces the need for informatics-driven integration of data, models, and governance, not technology alone (1)

3. **Critically analyze the structure of a modern healthcare system. (5). A. Identify the key actors involved in healthcare delivery and explain how their roles and interactions influence patient care outcomes (3). B. Explain the role of health insurance providers in this system and discuss the consequences for patients and providers when coordination between actors breaks down. (2)**

Ans.

- A modern healthcare system is a complex socio-technical system composed of interacting clinical, administrative, financial, and regulatory subsystems (1)
- Care delivery involves multiple interdependent actors, rather than a single decision-maker (1)
- Decisions are distributed across different levels (patient, provider, organization, payer) (1)
- The system operates under constraints of incentives, resources, and regulations (1)
- Patient outcomes depend on coordination and alignment between system components, not only clinical expertise (1)

A. Clinicians: diagnose, treat, and make clinical decisions directly affecting patient outcomes (1)

Patients: influence outcomes through preferences, adherence, and engagement in care (1)

Hospitals, laboratories, pharmacies, and insurers: provide infrastructure, diagnostics, medications, and financial access that shape care quality and timeliness (1)

B. Health insurance providers shape care through coverage policies, reimbursement rules, and incentives that influence clinical decisions and access to services (1)

When coordination between insurers, providers, and patients breaks down, consequences include delayed or denied care, increased costs, inefficiency, and provider burnout (1)

- 4. A researcher publishes ICU patient data, including names, admission dates, and rare diagnoses.(2*5)**
A. Identify the major privacy risk involved. B. Explain why patients with rare diseases are especially vulnerable.C. State one effect this could have on future clinical research. D. Mention one HIPAA-compliant method that would allow data sharing. E. State one ethical responsibility of researchers when handling sensitive ICU patient data.

Ans.

- A. The major privacy risk is patient identification, because names and admission dates are shared (1)
If patients are identified, their personal medical information becomes public, leading to possible harm or misuse (1)
- B. Rare diseases affect very few people, so identifying one patient becomes easier (1)
Once identified, patients may face stigma, discrimination, or unwanted attention (1)
- C. Patients may lose trust and refuse to share data or participate in studies, which can slow down medical research (2)
- D. **De-identification (Safe Harbor method)**, where names, dates, and identifying details are removed before data sharing (2)
- E. Researchers must prevent patient identification and protect confidentiality to avoid harm and respect patient rights (2)